



BUSINESS ANALYSIS

Lecture 6: Different Types of Requirements

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WHAT ARE REQUIREMENTS?

- “(1) A condition or capability needed by a user to solve a problem or achieve an objective.
- (2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification or other formally imposed documents.
- (3) A documented representation of a condition or capability as in (1) or (2).”

Source: IBM; <https://www.ibm.com/think/topics/what-is-requirements-management>

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DIFFERENT TYPES OF REQUIREMENTS

- **“Business requirements**

These requirements are the expectations, metrics and business needs that must be met for the system to accomplish the organization’s goals.

- **System requirements**

Also referred to as technical requirements, these requirements include details about the underlying technology that are vital to the success of the deliverable.

- **User or stakeholder requirements**

These requirements define the features and functions of a system necessary for satisfying end user expectations and needs.”

Source: IBM; <https://www.ibm.com/think/topics/what-is-requirements-management>

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DIFFERENT CATEGORIES OF REQUIREMENTS

- **“Functional requirements**

Functional requirements refer to the functionality of a new system, product or application. These requirements are measurable and critical to meeting the expectations and needs of the end users.

Example: The software should allow users to sign in with their IBM accounts..

- **Non-functional requirements**

Non-functional requirements refer to the performance of a system, product or application’s functional requirements. For instance, a system’s security, reliability and usability are all nonfunctional requirements.

Example: The online banking portal should fully load within three seconds after login.

- **Domain requirements**

Domain requirements refer to specific regulations, standards, rules and terminologies within an industry or domain that the system, product or application must adhere to.

Example: The healthcare application must comply with Health Insurance Portability and Accountability Act (HIPAA) requirements.”

Source: IBM; <https://www.ibm.com/think/topics/what-is-requirements-management>

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ACTIVITY: REQUIREMENTS ENGINEERING CASE

Now that you know

- What is requirements engineering
- Why it is necessary
- The steps involved in requirements engineering
- What is stakeholder management
- Why is stakeholder management important
- How to prioritize stakeholders through the Power-Interest Grid
- What is a stakeholder communication plan
- What are the different types of requirements

Let us start working on a project

Source: See reference/reading list at the end of the presentation

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ACTIVITY: REQUIREMENTS ENGINEERING PROJECT

- Identify a business/company of interest (imaginary or actual)
- Conduct SWOT analysis and **identify a key requirement (think of it as a project, i.e., it should have a deliverable, a plan, and is time and cost bound)**
- Consult the Linear Requirements Engineering Model for mapping that requirement
- Prioritize stakeholders through the Power-Interest Grid
- Design a stakeholder communication plan
- Identify business requirements, stakeholder requirements and system requirements
- Identify functional requirements, non-functional requirements and domain requirements

Think scientifically and creatively about your project: How do business analysts resolve issues?

Example 1: adept; <https://adept-plm.com/about-adept/clients/>

Example 2: AISIGHT.ai; <https://www.aisight.ai/>

Source: adept; <https://adept-plm.com/about-adept/clients/>
 AISIGHT.ai; <https://www.aisight.ai/>

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ESSENTIAL READINGS

IBM: "What is requirements management", available at <https://www.ibm.com/think/topics/what-is-requirements-management>

adept: "Product line system management"; available at <https://adept-plm.com/about-adept/clients/>

AI SiGHT.ai: "Make your market share data actionable"; available at <https://www.aisight.ai/>

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REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK);** IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)

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